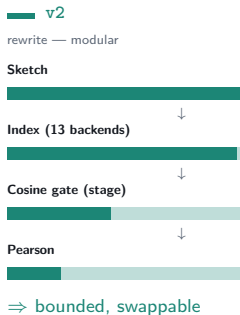
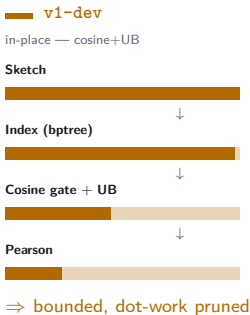
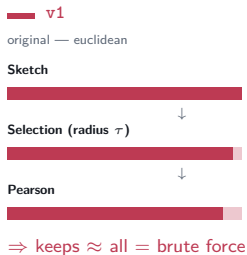


CorrTrack — three lineages, one pipeline

Share of pairs passing each stage (dense data). The funnel that narrows = a real filter.



thesis **v1** (radius, unbounded) doesn't narrow → Pearson runs on everything. **v1-dev** and **v2** bound it with a cosine gate — by opposite routes.

Overview — v1 → v1-dev → v2

	v1 original	v1-dev in-place	v2 rewrite
Sketch	Rademacher ± 1 RP + toggle	same RP, unit-norm for cosine	Gaussian + 5 methods , swappable
Index	bptree/flat, 1D key	bptree + UB-prune infra	13 indexes , cdist batch
Selection	euclidean radius (unbounded)	cosine γ fused + UB, $ \cos $	cosine gate stage, cdist, $ \cos $
Validation	Pearson + kurtosis + min_dist	+ hybrid adaptive skip	correlate, std only, signed lag
Monitoring	in/out, Cython kernel	+ numeric fast path	Monitor, pure bookkeeping

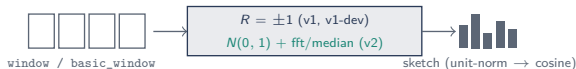
v1 → v1-dev in-place: euclidean → **cosine gate** fused in the Cython kernel + Cauchy–Schwarz UB pruning, single-insert.

v1 → v2 modular rewrite: separate cosine stage, 13 swappable indexes, backend-swappable compute, 5 sketch methods.

v1-dev $\approx$ v2 same selection idea (cosine, single-insert) — v1-dev **deep-optimizes the monolith**, v2 **modularizes**.

Step 1 — Sketch: projecting the window

Role — turn each window into a short vector ($n_vectors$) that preserves correlation (JL projection).



■ v1

original

■ v1-dev

in-place

■ v2

rewrite

- **Rademacher** ± 1 + toggle
- **same** RP + toggle
- Gaussian $N(0, 1)$ + znormalize
- norm z | mean_12, Cython
- **unit-norm** $\rightarrow$ dot = cosine
- **5 methods**: RP, fft, median

v1 $\rightarrow$ v1-dev same projection; normalization becomes unit (ℓ_2) so the candidate dot equals the cosine.

v1-dev $\rightarrow$ v2 same RP idea; v2 adds sketch *variety* (FFT/median = AUC lever) + swappable backend.

Step 2 — Index / neighbor retrieval

Role — for each window, find its neighbors in sketch space (raw candidates).



v1 / v1-dev: 1D key $[v_0 \pm \tau]$ + UB-prune



v2: L2 ball, query_radius

— v1

original

- bptree/flat, 1D key
- range + L2 refine

— v1-dev

in-place

- bptree (AVL), 1D key
- + **UB-prune** (skip dots)

— v2

rewrite

- **13 indexes** (grid/bst/hnsw...)
- BLAS cdist batch

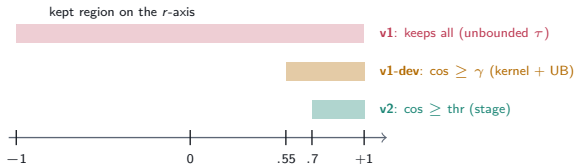
v1 → v1-dev

same bptree; the scan gains a Cauchy–Schwarz **upper bound** to skip full dot products.

v1-dev → v2

v1-dev **prunes work** in one smart tree; v2 **swaps** among many brute-but-vectorized indexes.

Step 3 — Selection / cosine gate: what each keeps



v1

original

- euclidean cell_size (τ)
- $r \gtrsim 1 - \tau^2/2$, **unbounded**
- double-insert - vec

v1-dev

in-place

- cosine γ fused in kernel
- $\gamma = 1 - \tau^2/2$ (overridable) + UB
- single-insert $|\cos|$

v2

rewrite

- cosine gate *separate* stage
- sketch_threshold, cdist
- single-insert $|\cos|$

$v1 \rightarrow v1\text{-dev}/v2$ unbounded radius $\rightarrow$ **bounded cosine** $\rightarrow$ $v1$'s "keep everything" becomes inexpressible.

shared limit $\cos \approx r$ is noisy: recall $0.95 \Rightarrow \gamma \approx 0.55$ (not 0.7) $\Rightarrow$ wide band $[0.55, 0.7)$ for both.

Step 4 — Exact Pearson validation

Role — exact correlation on raw windows: confirm or reject each candidate.



v1

original

- `fast_corr_and_dist`
- `std < 10-3`, `kurt > 5`, `min_dist`

v1-dev

in-place

- **same** Pearson + guards
- + **hybrid** skip (EMA repeat)

v2

rewrite

- `backend.correlate` (swappable)
- `std_threshold`, signed lag

v1 → v1-dev

same exact Pearson; a **hybrid** layer skips re-validating stable, repeating pairs.

v1-dev → v2

v1-dev cuts validation *count* (hybrid); v2 cuts *cost/pair* (vectorized swappable backend).

Step 5 — Persistence monitoring

Role — track over time the lifetime of correlations and their anomalies (onset, drop, sign change).



— v1

original

- `corr_lengths/anomalies`
- `_in_corr/_out_corr`, Cython

— v1-dev

in-place

- **same** model (in/out)
- + **numeric fast path**

— v2

rewrite

- Monitor class, typed anomalies
- **pure bookkeeping**, no backend

v1 → v1-dev

same lifetime model; a numeric fast path added to the existing Cython monitoring.

v1-dev → v2

v1-dev keeps the coupled Cython+spill design; v2 **decouples** it into pure state.